



HARIT PATHSALA — हरित पाठशाला

Can We Trust the Eco-Score?

A mathematical audit of the number we show students — and why 9.19% of a school was being told their effort was worth nothing.

Jayed Alam Mansur

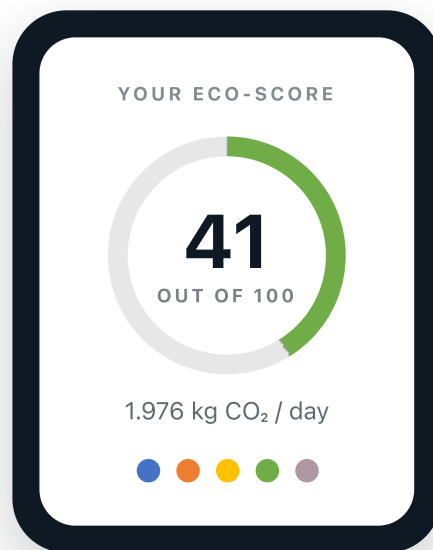
B.Tech Artificial Intelligence, Year 3 · Kathmandu University

Supervised by [Mathematics Teacher] · Sandesh Thakuri

Where This Started

Harit Pathsala began at the Nepal Climate Hackathon 2025 as a carbon-footprint calculator for school students. Five questions, one number.

A student tells it how they get to school, how much electricity the house used last month, how many cooking-gas cylinders, how much waste, how much they spend on stationery. It adds those up into kilograms of CO₂ per day and converts that into a single score out of 100 — the Eco-Score.



That is the whole product, and it works. The student above — bus, 2.0 km each way, 90 kWh a month, one cylinder, medium waste, Rs 200 of stationery — emits **1.976 kg CO₂/day** and scores **41**.

The number stopped being a readout and started being a judgement. Teachers show it to students. Students compare it with each other. A dashboard averages it across a class. Nobody had ever checked whether it deserved that trust.

The observation this research came from

This is not a complaint about the app. It is the audit the app never got. Everything that follows separates two things that are easy to confuse: the **footprint calculation**, which is physics and is correct, and the **scoring function**, which is a design decision and is not.

Three Problems Hidden Inside the Formula

Before any statistics, three structural problems are visible from the shape of the scoring rule alone.

The rule is a straight line between two chosen boundaries: 0.5 kg CO₂/day scores 100, 3.0 kg scores 0, and anything outside is clamped to the nearest end. Three consequences follow before a single student is scored.

1

A floor that swallows people

Above 3.0 kg/day everyone scores 0 — the student just over the line and the heaviest emitter in the school, reported identically.

2

A ceiling nobody reaches

The top of the scale is unreachable in practice, so the reward end of the range is wasted and the score can never say "excellent".

3

Boundaries from the wrong scale

0.5 and 3.0 were chosen a priori, not derived from what these emission factors actually produce on a Nepali household.

Each of the three is a claim, and each is tested later with a number: problem one in chapter 7, problem two in the same place, problem three by the household baseline. The point of stating them now is that they are all visible from the geometry — a flat line carries no information, whatever the data.

Building the Model

Five activities, five constants. Every constant is a physical derivation, not a guess — which is exactly what the first question from my supervisor asked about.

● **Transport — 0.016 / 0.066 / 0.19 kg CO₂ per km**

Fuel emission per litre, divided by the vehicle's efficiency, divided again by how many people share the ride. A bus burns more fuel than a motorbike but splits it forty ways, which is why it lands an order of magnitude lower per passenger.

Bus 0.016 (4 km/L ÷ 40 passengers) · motorbike 0.066 (35 km/L) · car 0.19 (12 km/L). Walking and cycling are 0. Doubled in the formula for the return trip.

● **Electricity — 0.23 kg CO₂ per kWh**

Nepal's grid is roughly 82% hydro and 18% thermal, which makes a unit of Nepali electricity unusually clean by world standards.

NEA Annual Report 2022/23. Rooftop solar uses 0.04 kg CO₂/kWh, a lifecycle estimate.

● **Cooking — 21.442 kg CO₂ per cylinder**

A Nepali LPG cylinder holds 14.2 kg of gas. Burning one kilogram releases 1.51 kg of CO₂ — the carbon content multiplied by 44/12, the mass ratio of CO₂ to carbon.

IPCC 2006 Guidelines, Volume 2. $14.2 \times 1.51 = 21.442$.

● **Waste — 0.827 kg CO₂e per kg**

Waste in a landfill decomposes into methane and CO₂. Methane is the more damaging gas, so the total is expressed as CO₂-equivalent.

IPCC 2006 Guidelines, Volume 5 (Waste).

● **Stationery — 0.0017285 kg CO₂e per NPR**

The only spend-based factor: emissions per rupee spent, covering the whole supply chain behind a notebook. It is also the least certain of the five.

GHG Protocol spend-based method — 1.7285 kg CO₂e per 1000 NPR.

TEACHER'S QUESTION 1

What were the fixed constants for each activity?

The five numbers above, and they are fixed in the strict sense: not one of them was fitted, tuned or adjusted anywhere in this research. They come from the hackathon organiser's *Emission_factors.xlsx*, are mirrored in

src/logic.js, and the notebook re-derives the documented worked example from them on every run — halting with an error if any category disagrees to four decimal places. The audit questions the scale that turns kilograms into a score; it never questions the kilograms.

From physics to a linear sum

Each activity contributes its own amount multiplied by its constant, and the daily footprint is just the sum of the five. Monthly quantities are divided by 30 and the commute is doubled for the return trip, so those constants are folded in:

$$F = \sum \theta_i x_i \quad (i = 1 \dots 5)$$

$$F = \theta_{\text{mode}} \cdot d \cdot 2 + (E/30) \cdot \theta_{\text{elec}} + (L \cdot \theta_{\text{LPG}})/30 + w \cdot \theta_{\text{waste}} + (P/30) \cdot \theta_{\text{stat}}$$

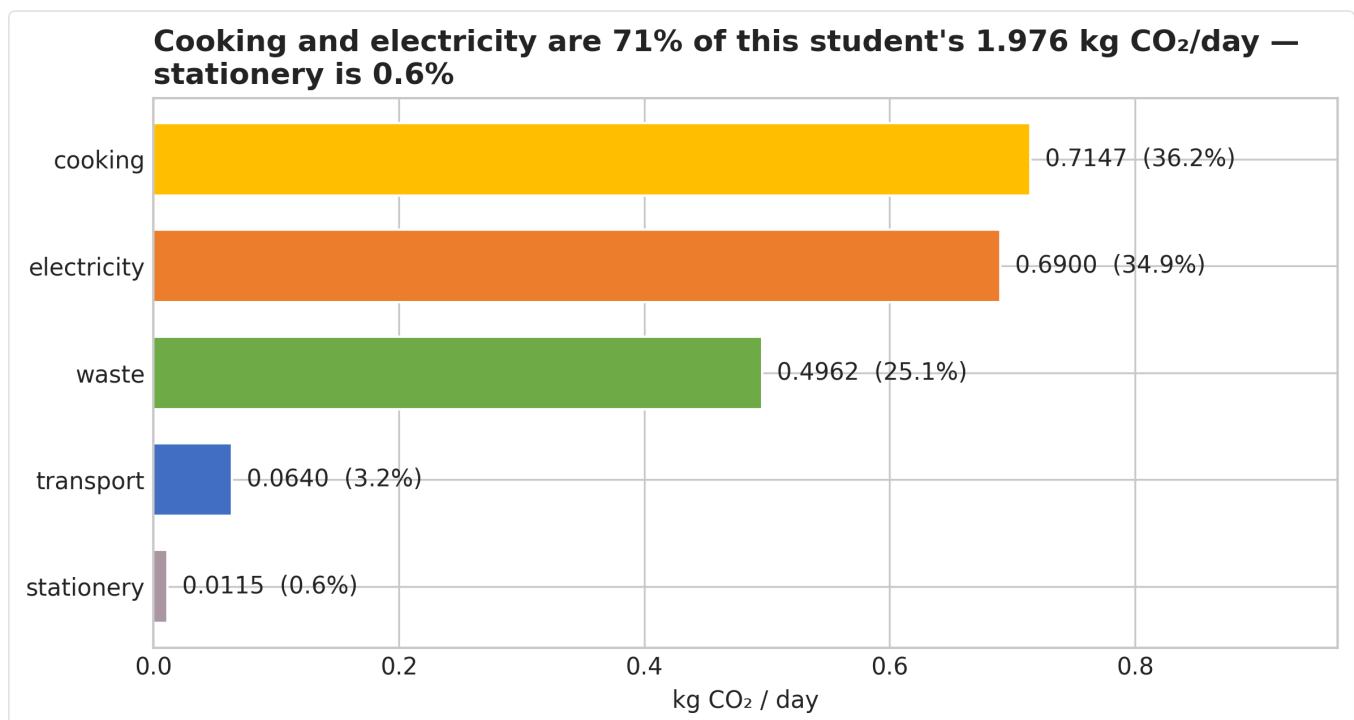
Linearity is not a convenience here, it is what makes the rest of the research possible. Because F is linear in the activities, each derivative is a constant. Because it is also linear in the emission factors, uncertainty propagates through an exact formula rather than an approximation. Both facts are used in chapters 4 and 5.

The second half of the model is the scoring step — and this is the part under investigation:

$$S(F) = \text{clamp}(0, 100, 100 - ((F - 0.5) / 2.5) \times 100)$$

$$dS/dF = -40 \text{ for } 0.5 \leq F \leq 3.0 \leftarrow \text{the score responds}$$

$$dS/dF = 0 \text{ otherwise} \leftarrow \text{the score is deaf}$$



The worked example broken into its five parts. Cooking and electricity are most of it; stationery is 0.6%.

From Formula to Simulation

The calculator treats its constants as facts. They are estimates — and once you admit that, the score stops being one number and becomes a distribution.

TEACHER'S QUESTION 2

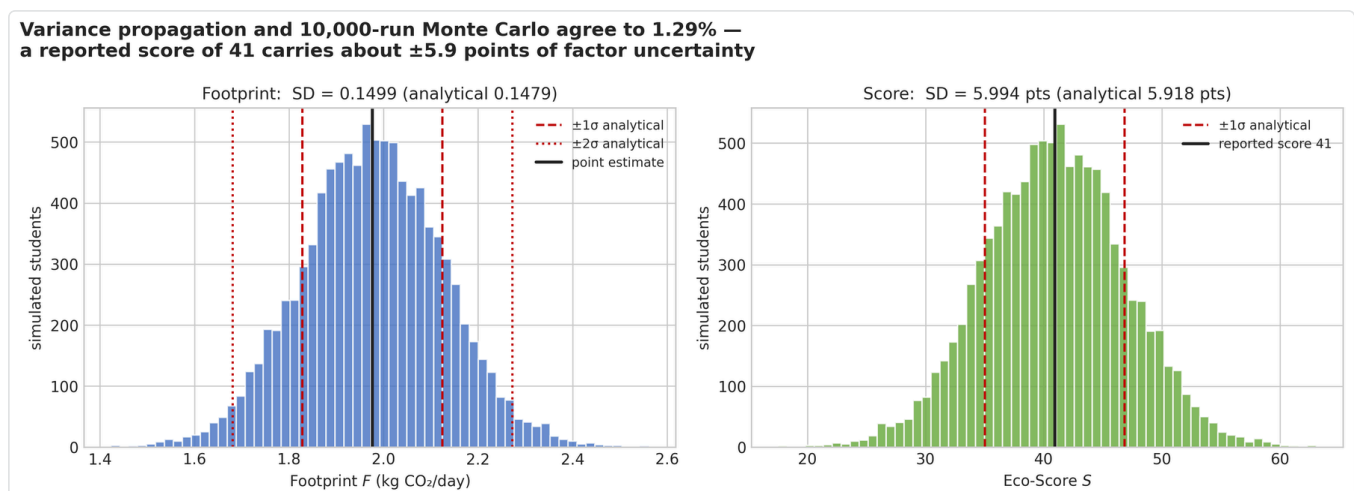
Which model was used before? Why Monte Carlo now?

Before this research the model was purely **deterministic**. You put five answers in, the five constants were treated as exact, and one score came out. Run it a thousand times with the same answers and you get the same score a thousand times. There was no notion that the score could have been different.

That framing hides something real. The figure **0.23** kg CO₂ per kWh is not a constant like π ; it is a ratio the Nepal Electricity Authority recomputes every year, and it moves as the hydro-to-thermal mix shifts with rainfall and imports. Across their 2018–2023 reports that swing supports about **±15%**. Waste composition varies about **±20%** between households; the spend-based stationery factor about **±25%**; the LPG fill weight about **±5%**; bus passenger loads about **±10%**.

So each constant becomes a random variable — a normal distribution centred on the published value with that spread, clipped to stay positive — and the question changes from "what is the score" to "how well do we know the score". Two independent methods answer it. Algebraically, because F is a linear sum, the variances simply add, giving a standard deviation of **5.92 points**. By simulation — 10,000 versions of the world, each with slightly different constants, the same student scored in every one — **5.99 points**. They agree to **1.29%**.

In plain terms, a reported score of 41 is really somewhere around 29 to 53. Two students less than about twelve points apart cannot honestly be ranked against each other.



The closed-form answer and the 10,000-run simulation land on the same spread — which is how you know neither is a mistake.

TEACHER'S QUESTION 3

What other methods could we have used?

Monte Carlo was not the only option, and it was not chosen for novelty. Four alternatives were considered and rejected for specific reasons:

METHOD	VERDICT	REASON
Interval arithmetic	Rejected	Propagates worst-case bounds, so every factor is assumed wrong in the same direction at once — a range too pessimistic to be useful.
Latin hypercube sampling	Not needed	Better sample coverage per draw, but with only five inputs and 10,000 draws plain Monte Carlo has already converged.
Bootstrap resampling	Impossible	Resamples observed data to estimate its spread — and there is no observed student data to resample yet.
Bayesian posterior (MCMC)	Overkill	Would need defensible priors on every emission factor and yields little beyond the variance we can already compute exactly.

The honest summary is that inside the responsive region the algebra alone is enough, and the simulation only confirms it. What earns Monte Carlo its place is the boundary — described at the end of this chapter.

TEACHER'S QUESTION 4

Is this the same Monte Carlo as in DAA?

Same name, opposite purpose. In Design & Analysis of Algorithms, a Monte Carlo algorithm uses randomness as a *computational trick* on a problem that is completely deterministic. Whether 561 is prime is a fixed fact; Miller–Rabin flips coins only to reach that fact faster than trial division, accepting a bounded chance of being wrong. Randomised QuickSort picks a random pivot for the same reason — the sorted order was never in doubt. The randomness lives in the method, never in the problem.

Here the randomness *is* the problem. The grid factor genuinely differs from year to year; that variation is a fact about the physical world, not about our algorithm. We are not approximating a number we could have computed exactly — we are characterising a spread that really exists. The two techniques share a name and a random number generator, and nothing else.

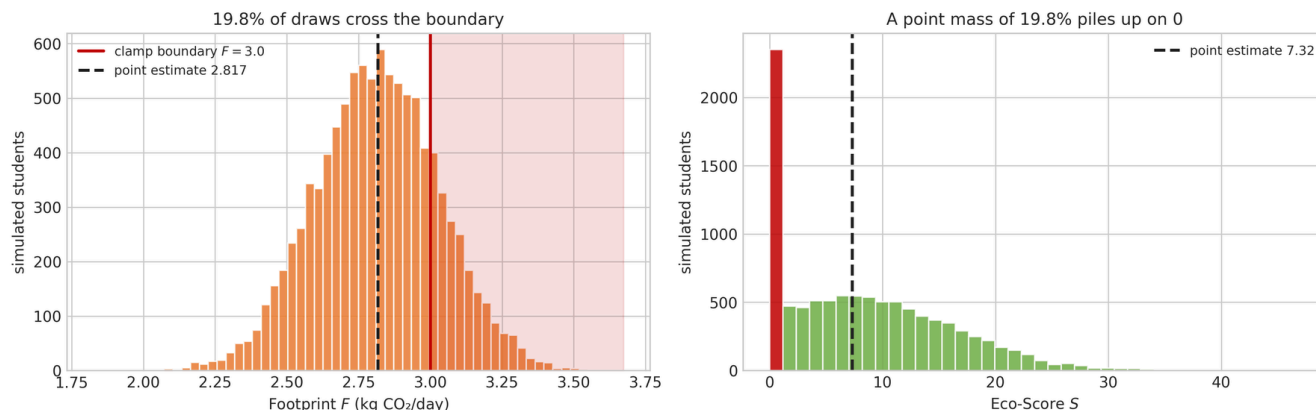
The moment the algebra stops working

There is one place where simulation is not a confirmation but the only correct tool, and it is the sharpest result in this chapter.

Take a student sitting just below the floor — a motorbike commute, a modest electricity bill, heavy cooking and heavy waste, adding to **2.817 kg CO₂/day**. The deterministic model reports a score of 7 and says nothing more. But the clamp is not a linear function, so the variance formula quietly stops describing what the app displays: part of the distribution is pushed past 3.0 kg and squashed onto the boundary. The mean shifts, the shape turns asymmetric, and a spike of probability appears at exactly zero.

Simulating it shows how much: **19.6%** of plausible worlds report this student's score as **0** rather than 7. Not because the student changed — because the emission factors are estimates and the clamp turns a small measurement wobble into a total loss of information. No closed-form standard deviation can tell you that. This is what Monte Carlo was for.

Near the boundary the analytical formula stops describing the reported score: the true distribution is asymmetric with 19.8% of its mass collapsed onto zero



A student at 2.817 kg CO₂/day. Left: the draws crossing the 3.0 boundary. Right: the point mass piling up on a reported score of zero.

Why the Habit You Cut Matters More Than You'd Think

If a student wants a better score, what should they actually change? The answer is not what the app's five equally-weighted questions imply.

This is a sensitivity analysis, and the method is a partial derivative: increase one habit, hold the other four still, and see how far the score moves. Applying the chain rule to the two halves of the model gives a result that is unusually clean:

$$\partial S / \partial x_i = (dS/dF) \cdot (\partial F / \partial x_i) = -40 \cdot \theta_i$$

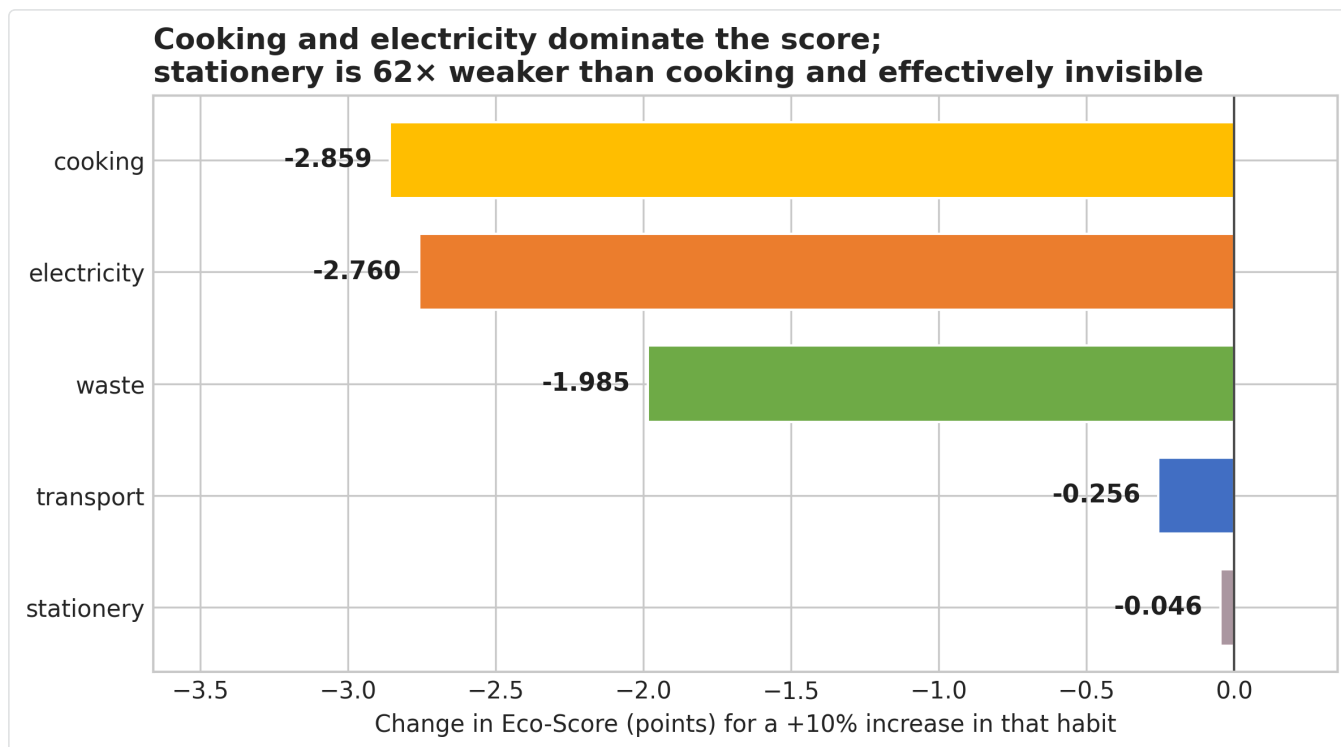
... inside the responsive region, and exactly 0 outside it.

Because the footprint is a weighted sum, the derivative of the score with respect to any habit is just -40 times that habit's emission factor. The ranking of what matters is therefore fixed by the constants alone and is identical for every student in the country.

Expressed as the score change from a realistic **+10% increase** in each habit — comparable across categories in a way raw derivatives are not, since kilometres, kilowatt-hours and rupees do not share units:

HABIT	SCORE CHANGE PER +10%	WHAT THAT MEANS
Cooking (LPG)	-2.859	one extra cylinder in ten months is visible
Electricity	-2.76	9 kWh a month is visible
Waste	-1.985	60 grams a day is visible
Transport	-0.256	400 metres a day is barely visible
Stationery	-0.046	Rs 20 a month is invisible

Cooking moves the score **62x** more than stationery. A student could stop buying notebooks and pens entirely, for a whole year, and move their score by less than a point. Any advice the app gives should be ordered by this table — not by which habit is easiest to talk about in a classroom.



The same five numbers as a tornado diagram. Two habits own the score; three barely register.

Two students the score cannot hear

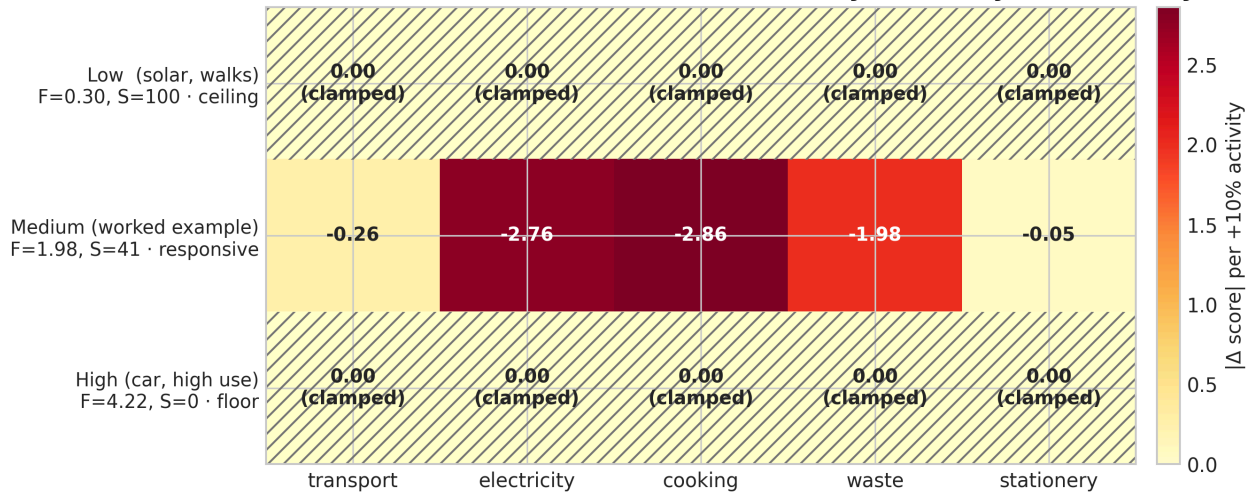
The second line of that derivative — *exactly 0 outside* — is easy to skim past and is the whole finding of this research. Consider two real archetypes.

A **very low-footprint student**: walks to school, rooftop solar, minimal cooking gas, careful with waste. Around 0.30 kg CO₂ a day. Their score is 100. If they somehow halved their footprint again, it would still be 100 — and the app cannot tell them how much better they are doing than another student also scoring 100 at three times their emissions.

A **very high-footprint student**: car commute, high electricity, heavy cooking and waste. Around 4.22 kg CO₂ a day. Their score is 0. If they halve their waste tomorrow, their score changes by **0.000 points**. Not a small amount — zero. In every one of the five categories, the derivative for this student is exactly nothing.

Both students are being told their effort is worthless, by a system that is functioning exactly as designed. The next chapter asks how many students that actually is.

**Only the middle student receives any feedback:
both dead-zone students have an elasticity of exactly zero in every category**



Three archetypes, five habits. Only the middle student receives any feedback at all; the hatched rows have an elasticity of exactly zero everywhere.

Building a Population of 10,000 Students

One worked example cannot tell you whether a scoring system works. You need a school.

TEACHER'S QUESTION 5

Why 10,000 synthetic students? What happened to the old data?

There is no old data. The Harit Pathsala calculator has been demonstrated and piloted, but no structured student survey had been collected when this analysis was written — there is no dataset of real answers sitting somewhere that I chose not to use. That is a genuine limitation, stated plainly rather than hidden, and it determines exactly what this chapter can and cannot prove.

The alternative to no data is not "wait indefinitely". It is to build a population whose distributions come from published Nepali sources, and then be precise about which conclusions depend on those distributions and which do not. Synthetic is not the same as fake: every marginal below is sourced, and crucially they were all fixed *before* any score was computed, so nothing could be tuned to produce a flattering answer.

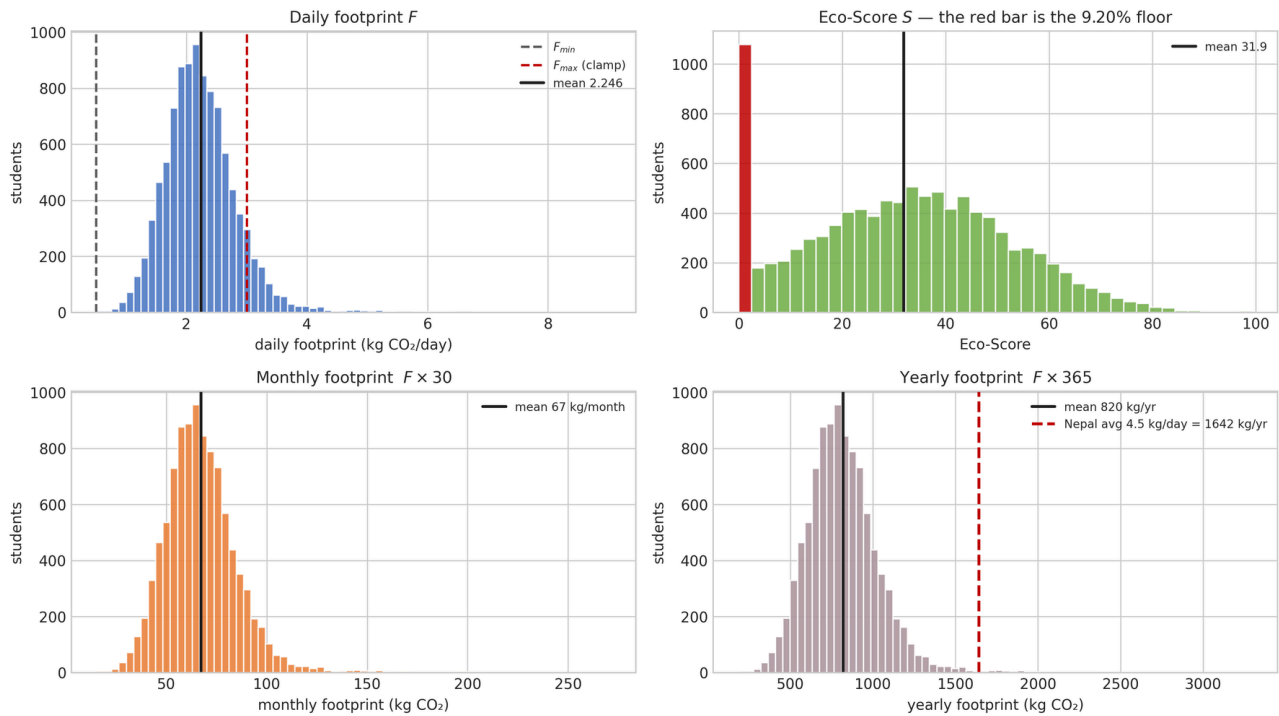
VARIABLE	DISTRIBUTION	SOURCE / REASONING
Transport mode	walk 0.2 · cycle 0.1 · bus 0.4 · motorbike 0.25 · car 0.05	Nepali student commute estimates
Commute distance	log-normal, mean 3.0 km, $\sigma=0.8$, clipped 0.5–20.0	commute distances are right-skewed — many near, a few far
Electricity	Normal(90.0, 35.0) kWh/mo, clipped 10.0–300.0	NEA domestic billing range
Rooftop solar	Bernoulli(0.05)	roughly 5% household penetration
LPG	Normal(1.2, 0.4) cyl/mo, clipped 0.25–4.0	typical household consumption
Waste	[0.3, 0.6, 1.2] kg/day, p = [0.3, 0.5, 0.2]	the app's own three answer options
Stationery	Normal(200.0, 80.0) NPR/mo, clipped 50.0–600.0	typical monthly spend

One parameterisation deserves a note, because it is the easiest thing in this chapter to get silently wrong. A log-normal distribution is defined by parameters in log-space, so passing the desired average straight in gives you a distribution whose *median* is 3 km and whose mean is over 4 km. Solving for the right log-space parameter instead — $\mu = \ln(\text{mean}) - \sigma^2/2$ — makes the arithmetic mean come out at exactly the 3.0 km intended.

The result is **10,000** students, generated from a fixed seed (42) so that every number in this document can be reproduced exactly.

What this can and cannot establish. It cannot validate the emission factors, and it cannot tell you the true floor percentage in a real Nepali school — those are assumptions in, assumptions out. What it *can* expose is structural behaviour of the scoring function that no single example reveals: where the mass of a plausible school lands on the curve, and how much of it lands somewhere the curve has gone flat. That conclusion depends on the shape of the population, not on the exact parameters, and it survives moderate changes to any row in the table above.

A plausible Nepali school: mean 2.246 kg CO₂/day, mean Eco-Score 31.9 — and a spike of 9.20% pinned at zero



The simulated school: daily, monthly and yearly footprints, and the resulting spread of Eco-Scores.

What the Population Revealed

Scoring ten thousand students turned three theoretical problems into three measurements.

 **TEACHER'S QUESTION 6**

Comparative results — before vs after our research?

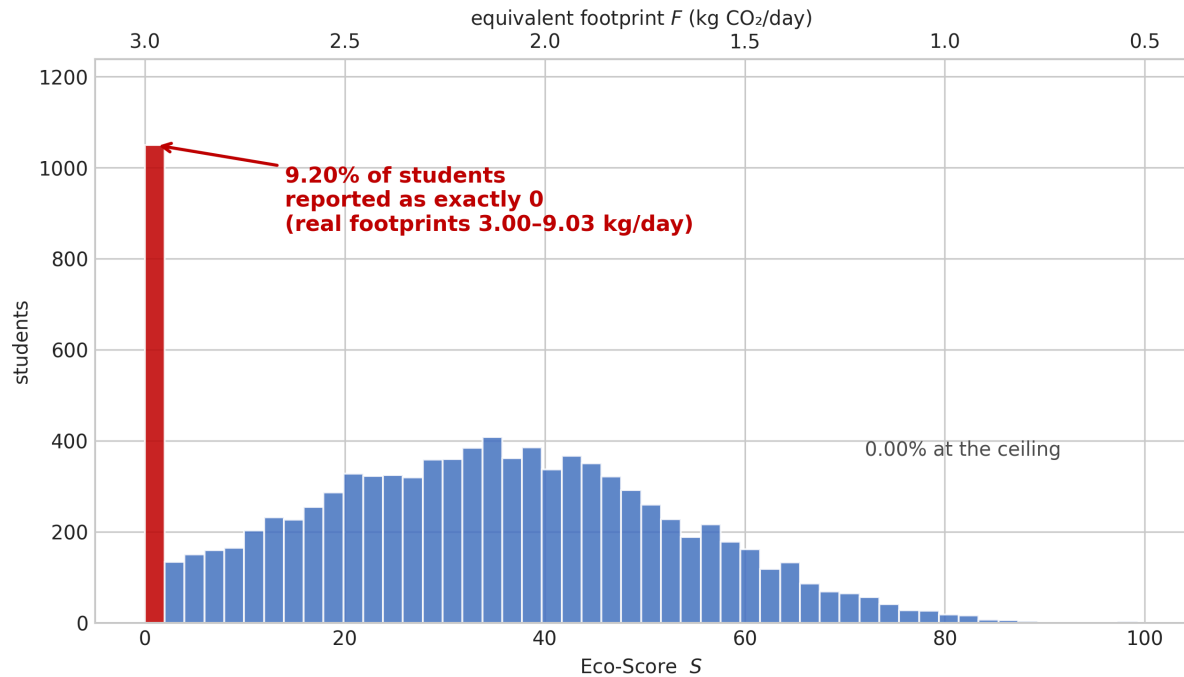
	BEFORE THIS RESEARCH	AFTER
Emission factors	assumed exact	random variables with quantified spread
Score precision	unknown — one number, no error bar	±5.92 points (1σ), confirmed two ways
Which habits matter	five questions, implicitly equal	ranked — a 62× spread from first to last
Behaviour across a school	never tested	tested on 10,000 students
Students given no feedback	unknown	9.19% scoring exactly 0
Diagnosis	—	formula sound; the clamp is the fault

The headline finding

Across the simulated school the mean footprint is **2.246 kg CO₂/day** and the mean Eco-Score is **31.9**. That average was not adjusted toward anything comfortable; it is what these emission factors produce on these distributions.

Then the floor. **9.19%** of the school scores exactly **0**, and **0.0%** reach 100 — not one student in ten thousand. The students at the floor are nothing like each other: their real footprints run from **2.99** to **8.91 kg CO₂/day**, a **3×** difference in actual emissions, reported as one identical number. And for every one of them the gradient is exactly zero, so no change they make will move it.

**9.20% of a plausible school is silenced by the floor —
a 3.0× range of real emissions collapsed into one number**



The red spike is 9.19% of the school, compressed onto a single value while their real emissions differ threefold.

Whose fault is it?

This is the question that decides whether the fix is a patch or a rewrite. Regressing the displayed score against the footprint across all ten thousand students gives **$R^2 = 0.8639$** — the score failing to reflect about 14% of what is happening. Removing only the students stuck at the floor and refitting gives **$R^2 = 0.9997$** .

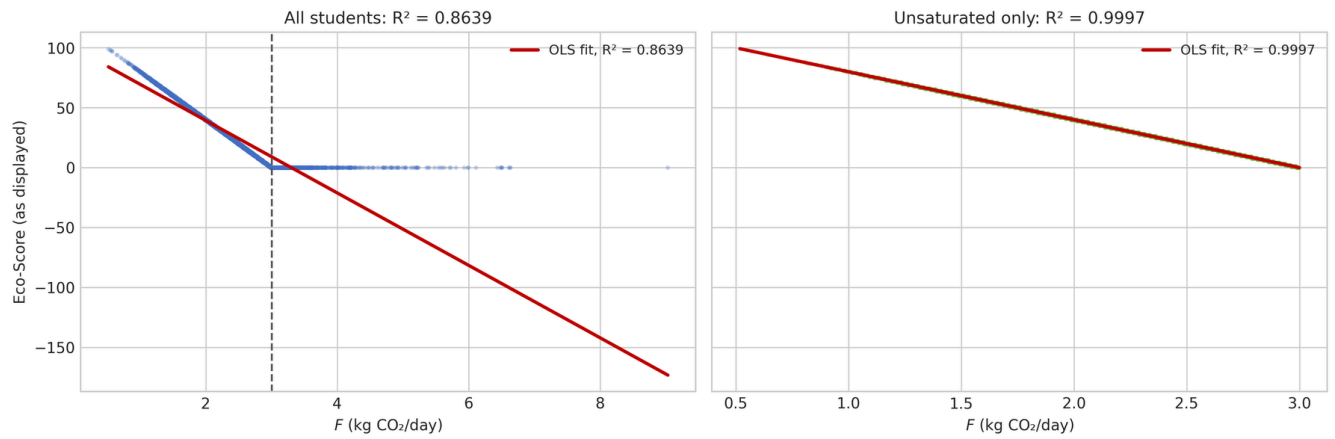
Among the students the system can actually score, it explains **99.97%** of the variation between them. The missing 0.03% is not model error either — it is precisely the variance of rounding the score to a whole number for display. There is nothing left to improve in the arithmetic.

The formula is not broken. The clamp is the entire problem.

The finding this research exists to establish

Chapter two's third problem can now be measured too. Electricity, cooking and waste alone — before a student travels a single kilometre or buys a single pen — average **2.048 kg CO₂/day**. The floor sits at 3.0. An ordinary student is therefore already 82% of the way to a score of zero because of household decisions they do not control. The boundaries were set for a scale these emission factors never produce.

The formula is not the problem: remove the clamped students and the score explains 99.97% of the variation between students



Left: all students, $R^2 = 0.8639$. Right: the clamped students removed, $R^2 = 0.9997$.

Fixing the Boundaries

If the clamp is the problem, replace the clamp — and nothing else. The physics stays exactly as it is.

Two replacements were derived and tested. Both leave F , all five emission factors and all five questions untouched; both change only the final step that maps kilograms to a score.

Method A — percentile boundaries

The simplest honest fix: stop inventing the endpoints and take them from the population. Setting the boundaries at the 5th and 95th percentile of the observed footprints gives **1.404** and **3.227 kg CO₂/day** in place of 0.5 and 3.0.

$$S(F) = \text{clamp}(0, 100, 100 - ((F - 1.404) / (3.227 - 1.404)) \times 100)$$

It spreads students across the range better than the original — its information content is the highest of the three, at **2.931 nats**. But the trade-off is fatal for our purpose: it still clamps. Ties reappear at both ends, and it now saturates **10.27%** of students — more than the original — because it adds a ceiling problem the original did not have.

Method B — the logistic curve (recommended)

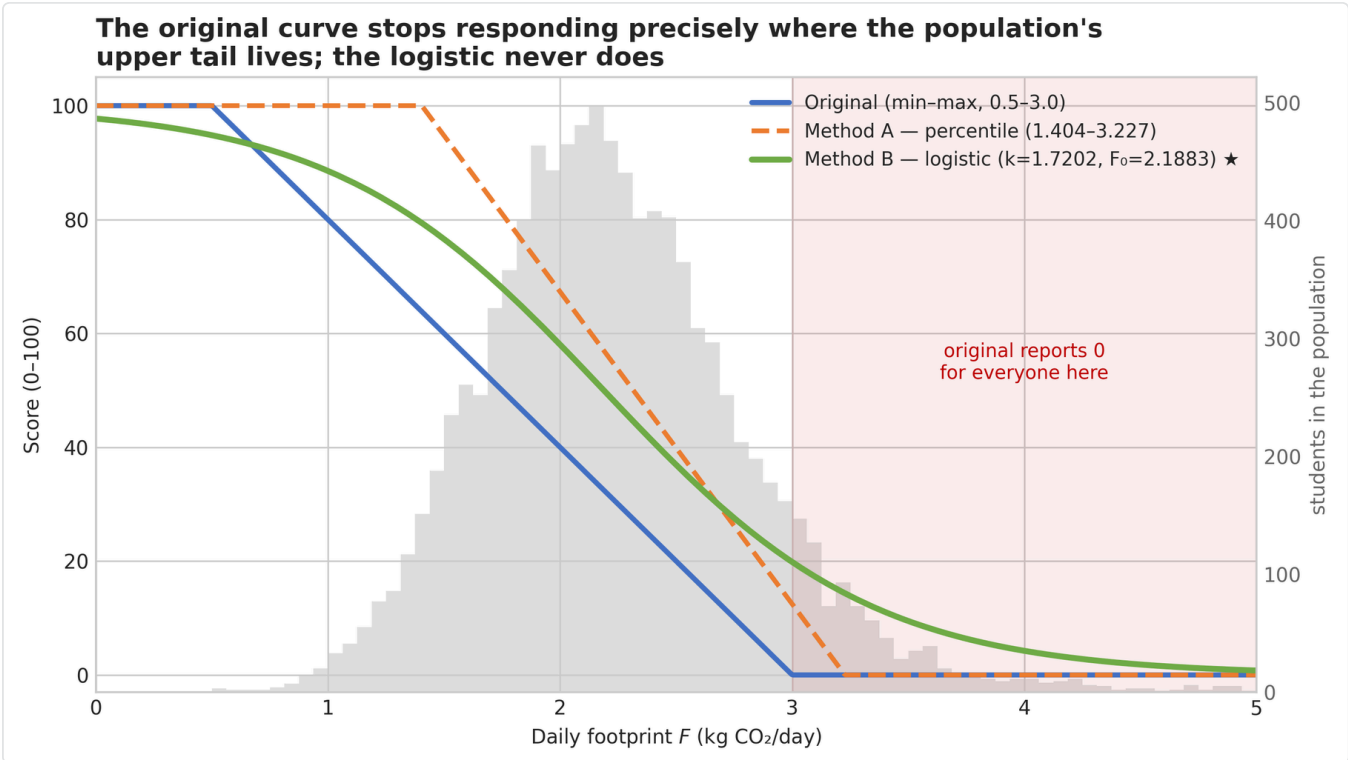
Instead of a straight line with two cliffs, an S-shaped curve that approaches 0 and 100 without ever arriving:

$$S(F) = 100 / (1 + e^{(k \cdot (F - F_0))})$$

$k = 1.7202 \leftarrow$ steepness, fitted to agree with the original score

$F_0 = 2.1883 \leftarrow$ the population median, which becomes the score-50 point

F_0 is the centre: the footprint that scores exactly 50, set to the median so that 50 means "a typical student". k controls how sharply the score falls, fitted by minimising the squared difference against the original score so the new scale agrees with the old one wherever the old one worked. The essential property is that the curve never goes flat — there is always somewhere left to go, in both directions.



The original stops responding exactly where the population's upper tail lives. The logistic never does.

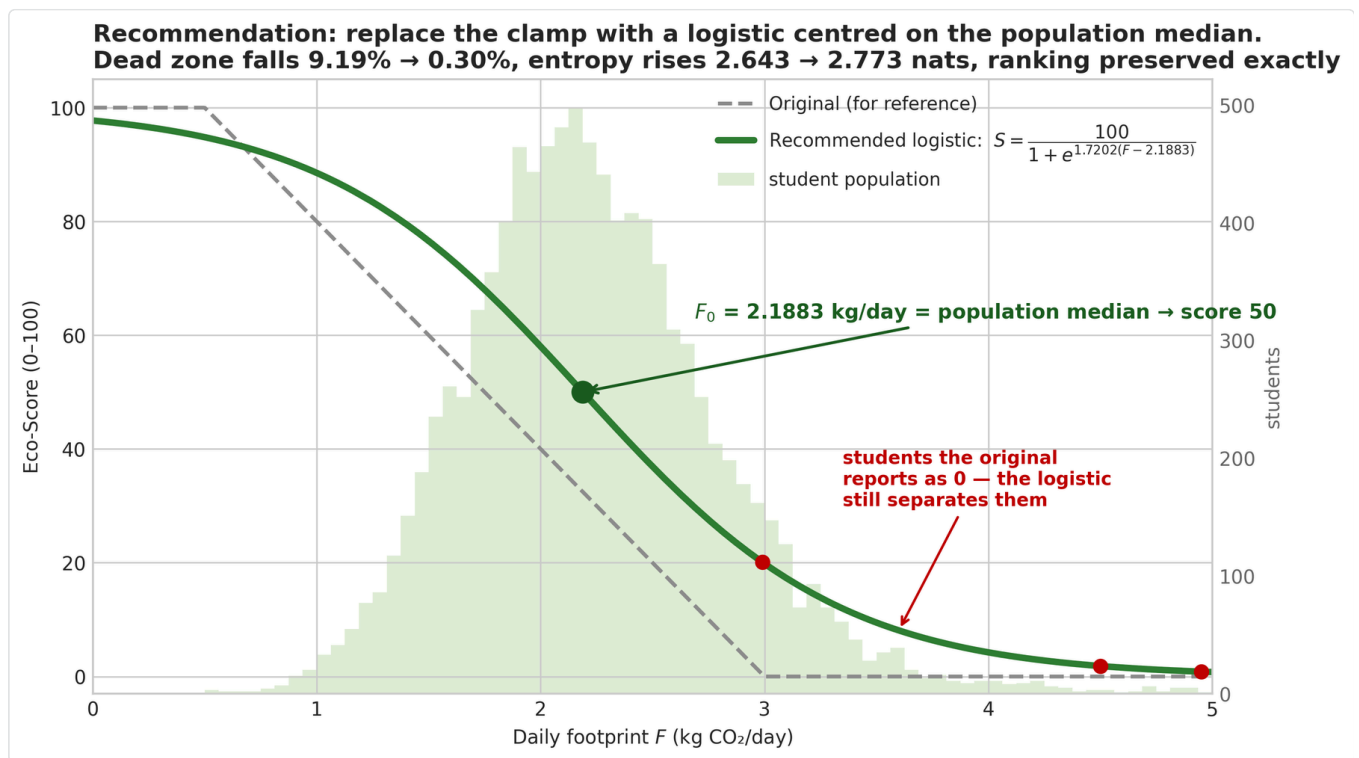
Side by side

CRITERION	ORIGINAL	PERCENTILE	LOGISTIC
Students given no feedback	9.19%	10.27%	0.3%
Information content (nats, max 2.996)	2.644	2.931	2.765
Preserves the true ranking	no — ties at 0	no — ties at both ends	yes, $p = 1.000000$
Agrees with the current score where it works	—	partly	$r = 0.9972$
Ever saturates	yes	yes	no

The decisive row is the third. A score exists to distinguish students, and the original ties hundreds of them at zero; the logistic preserves the footprint ordering *exactly*, with a Spearman correlation of 1.000000. If student A really emits less than student B, A scores higher — always. Meanwhile students who used to be silenced drop from **9.19%** to **0.3%**.

The trade-off, stated plainly. The logistic score is *population-relative*. Because F_0 is the median of the population it was fitted to, a score of 50 means "typical for this group", not "2.1883 kg CO₂ a day". Three consequences follow and none of them should be buried: the fit must be redone whenever the population changes; two schools cannot compare scores unless they pin the same k and F_0 ; and if everyone improves together, the average score does not move. That last one matters most for a tool meant to show progress — which is why the absolute kg CO₂/day figure

must stay on screen beside the score. Where an absolute threshold is what you need — "is this student under a sustainable limit?" — Method A with boundaries set from policy is the better instrument.



The recommended curve against the population it has to serve.

What Happens Next

A recalibration derived on synthetic data is a proposal, not a deployment. Five stages stand between here and a changed app.



TEACHER'S QUESTION 7

What is the full process after generating the data?

Generating the population was the middle of the work, not the end. The sequence from here is deliberately ordered so that nothing reaches a student until it has been checked against reality.

1 Collect real survey responses

Run the seven-question survey already specified in `data/survey_template.md` — three minutes, no personal data, wording provided in English and Nepali. Its CSV columns match the generator's output exactly, so real data drops straight into the notebook.

2 Refit the scale on that data

Recompute the percentile boundaries and re-fit k and F_0 on the real median. At least 100 responses are needed before a median is stable enough to anchor a scoring scale. The published constants then get replaced, and versioned.

3 Re-validate every finding

Re-run the whole notebook against real students and see which conclusions survive. The floor percentage will move. The existence of a flat region will not — that is structural.

4 Change one function in the app

Adopting the fix means editing the scoring function in `src/logic.js` and nothing else. No emission factor, no question, no footprint arithmetic changes. Show the kg CO₂/day figure beside the score so the absolute meaning is never lost.

5 Re-audit on a schedule

A population-relative scale drifts as its population changes. Refit annually, and repeat the saturation check each time — the failure this research found was invisible precisely because nobody looked twice.

WHAT STAYS

- All five emission factors, unchanged
- The footprint formula $F = \sum_i \theta_i x_i$
- The five questions students answer
- The daily kg CO₂ figure
- The existing app, UI and game

WHAT CHANGES

- One function: kilograms → score
- Advice ordered by real sensitivity
- Scores shown with their uncertainty
- A score that never stops responding
- Constants refitted as data arrives

Three Questions, Three Answers

Stripped of detail, the research asked three questions and each belongs to a different branch of mathematics.

Mathematical modelling — what *is* the model?

The footprint is a linear weighted sum of five activities, and the score is a clamped affine map of that sum. Writing it that way is not pedantry: linearity is what makes the derivatives constant and the variance exact, and the clamp is what makes the derivative vanish. Every later finding is a consequence of those two structural facts, which were invisible while the model lived only as code.

Sensitivity analysis — which inputs actually matter?

Partial derivatives, expressed as the score change from a +10% shift in each habit. Cooking at -2.859 points and electricity at -2.76 dominate; stationery at -0.046 is 62 times weaker than cooking and cannot meaningfully move anything. And in the clamped regions every one of those derivatives is exactly zero — which is how a sensitivity analysis found a design flaw rather than just a ranking.

Statistical calibration — is the output trustworthy?

Two answers. On precision: a single score carries ± 5.92 points of uncertainty from the emission factors alone, confirmed by simulation to within 1.29%. On calibration: the mapping saturates 9.19% of a plausible school, and replacing it with a logistic curve reduces that to 0.3% while preserving the ranking exactly.

What I am not claiming

Three limitations, stated deliberately. The population is **simulated, not surveyed**, so 9.19% is what these factors produce on a plausible school, not a measured fact about a real one. The logistic scale is **population-relative** and must be refitted when the population changes. And the emission factors were assumed to vary **independently** — they probably do not, and positive correlation would make the true uncertainty larger than 5.92 points, so that figure is a lower bound rather than a final answer.

MAIN FINDING

The formula was always correct. The boundaries were always the problem.

Harit Pathsala — Eco-Score research

Jayed Alam Mansur · B.Tech Artificial Intelligence, Year 3 · Kathmandu University

Supervised by [Mathematics Teacher] · Sandesh Thakuri

Every figure in this document is generated by `research/notebook/eco_score_research.ipynb` and every number is read from `research/data/key_results.json`. Population of 10,000 students, seed 42; 10,000 Monte Carlo draws; 14 figures at 300 DPI.

Emission factors from the Nepal Climate Hackathon 2025 organiser's `Emission_factors.xlsx`, via `src/logic.js`. IPCC 2006 Guidelines (Vol. 2 Energy, Vol. 5 Waste) · NEA Annual Report 2022/23 · GHG Protocol · UK DEFRA 2023 · Shannon (1948).